mHAVOK on Lorenz

Parallel event metrics, Chamfer distance, and channel-combo comparison

Task completed

- Standalone implementation extracted to `mhavok_lorenz.py`
- Full metrics exported for **recall, precision, accuracy, F1, mean linear R^2, and Chamfer distance**
- All single-, double-, and triple-channel combinations of `x`, `y`, and `z` benchmarked

Main conclusion

- For the default `x+z` setup, the best **alignment** model and best **event-F1** model are different.
- Across all channel combinations, the best event-detection model on the tested grid is **`x+y+z`**.

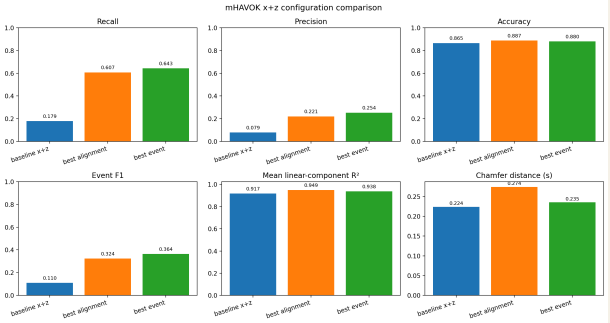
x+z Baseline vs Tuned mHAVOK Models

Baseline and winners

Model	Delays	Rank	Recall	Precision	Accuracy	F1	Mean linear R^2	Chamfer (s)
Baseline x+z	100	9	0.1786	0.0794	0.8653	0.1099	0.9173	0.2236
Best alignment	150	13	0.6071	0.2208	0.8872	0.3238	0.9493	0.2740
Best event model	150	11	0.6429	0.2535	0.8801	0.3636	0.9382	0.2352

Takeaway

- The tuned x+z model clearly improves event metrics over the baseline.
- The **best alignment** model is not the same as the **best event-**



Direct comparison of baseline, best-alignment, and best-event x+z models on the requested metrics.

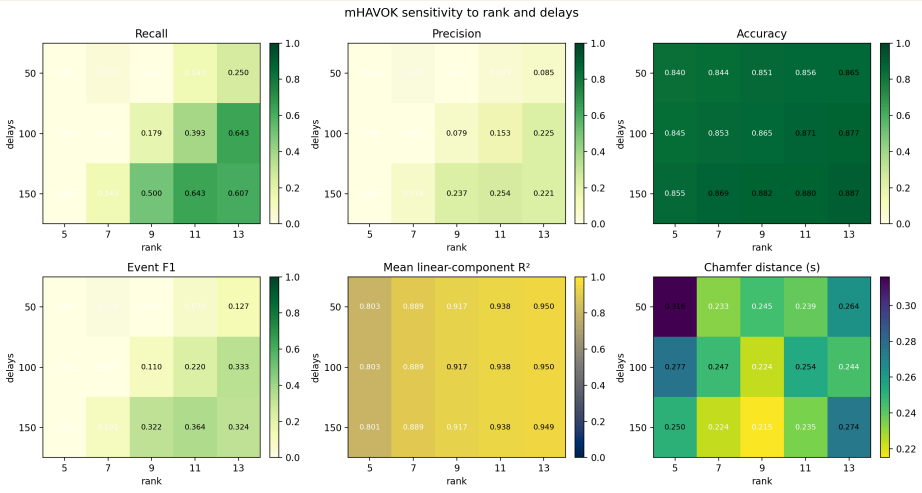
Rank-Delay Sweep for Event Metrics

Best x+z settings on the tested grid

Objective	Delays	Rank	Recall	Precision	Accuracy	F1	Mean linear R^2	Chamfer (s)
Best event F1	150	11	0.6429	0.2535	0.8801	0.3636	0.9382	0.2352
Best alignment	150	13	0.6071	0.2208	0.8872	0.3238	0.9493	0.2740

Reading the heatmap

- Event metrics improve strongly at **higher delays**.
- High mean linear R^2 does not uniquely determine the best event model.
- Chamfer distance and F1 do not select the same point as tight alignment.



Recall, precision, accuracy, F1, mean linear R^2, and Chamfer distance over the tested rank-delay grid.

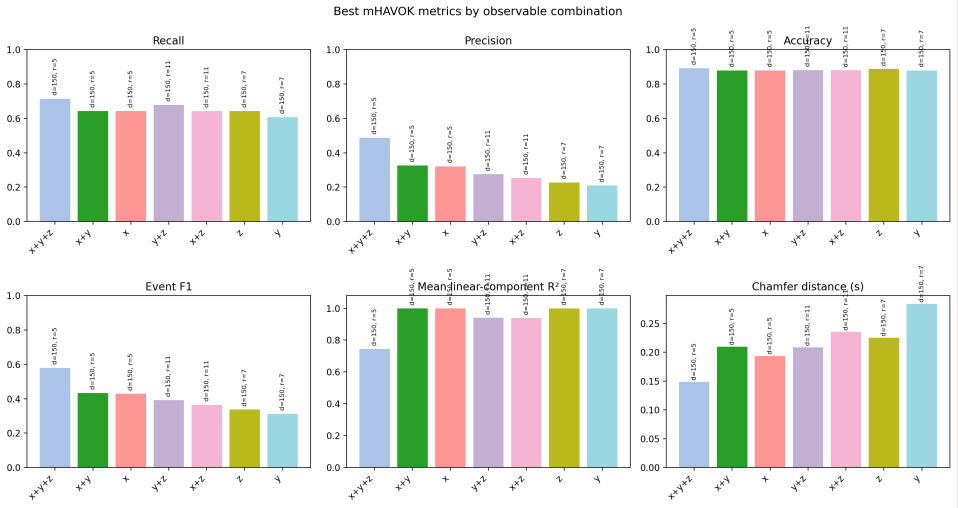
Channel Combinations: x, y, z Matter

Best model from each channel combination

Combo	Delays	Rank	Recall	Precision	Accuracy	F1	Mean linear R^2	Chamfer (s)
x+y+z	150	5	0.7143	0.4878	0.8924	0.5797	0.7450	0.1488
x+y	150	5	0.6429	0.3273	0.8794	0.4337	0.9988	0.2095
x	150	5	0.6429	0.3214	0.8792	0.4286	0.9995	0.1937
x+z	150	11	0.6429	0.2535	0.8801	0.3636	0.9382	0.2352

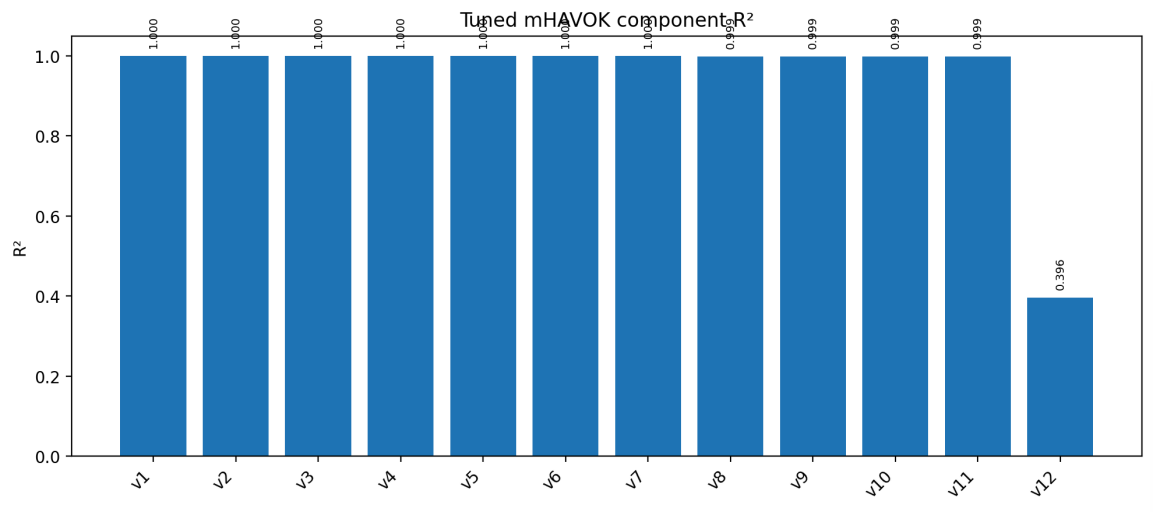
Main result

- The strongest event-detection model on the tested grid is **x+y+z**.
- Adding `y` materially changes the ranking.



Per-combo best recall, precision, accuracy, F1, mean linear R^2, and Chamfer distance.

R² Diagnostics and Objective Tradeoffs



Tuned x+z component R² values: most modes are near-perfect, but the last mode is much weaker.

Tuned x+z component fit

- `v1` through `v7`: **1.0000**
- `v8`: **0.9991**
- `v9`: **0.9989**
- `v10`: **0.9992**
- `v11`: **0.9986**
- `v12`: **0.3957**

Observable-selection winners from the smaller objective table

Objective	Winner
Tight alignment	z only
Balanced	x and z
Early warning	x only

Interpretation

- Objective choice matters even before the full channel-combo benchmark.

Final mHAVOK Takeaways

What is now in the code

- Standalone manual mHAVOK implementation in `mhavok_lorenz.py`
- Exported CSV summaries and slide-ready figures in `plots/mhavok_lorenz`
- Dedicated metrics for **recall, precision, accuracy, R^2, and Chamfer distance**

Scientific conclusion

- The best mHAVOK model depends on the objective.
- For the full channel benchmark, **x+y+z** is the best event-detection model on the tested grid.
- For the default x+z setup, **best alignment** and **best event F1** are different configurations.

BEST X+Z DELAYS, RANK

150, 11

BEST COMBO DELAYS, RANK

150, 5

BEST COMBO CHAMFER

0.149 s